

MSMES, PLI, SEMICONDUCTORS AND MANUFACTURING STRATEGY

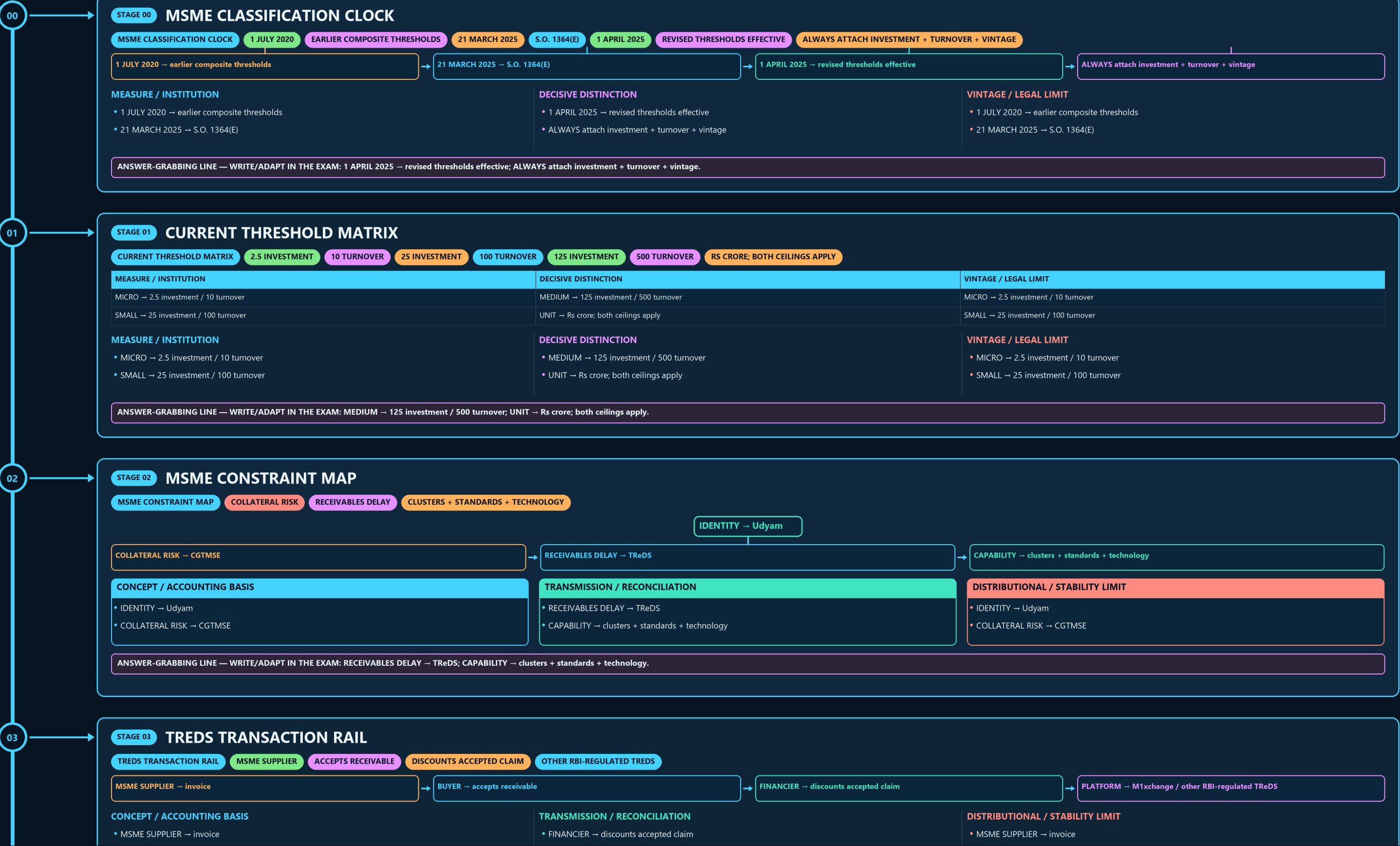
MSME CLASSIFICATION CLOCK → CURRENT THRESHOLD MATRIX → MSME CONSTRAINT MAP → TREDs TRANSACTION RAIL → PLI STATUS LADDER → PLI SCORECARD

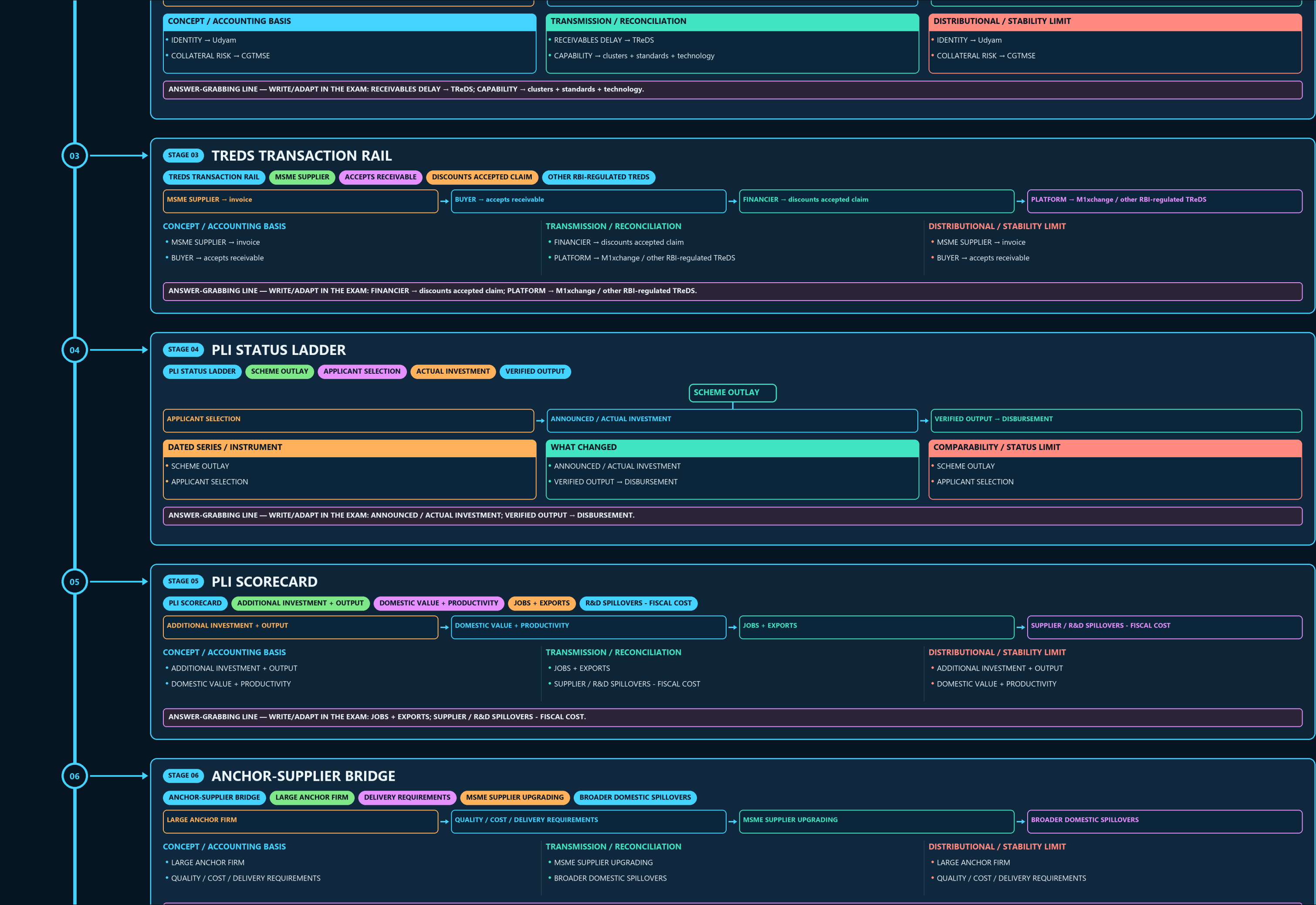
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





CONCEPT / ACCOUNTING BASIS

• ADDITIONAL INVESTMENT + OUTPUT

• DOMESTIC VALUE + PRODUCTIVITY

TRANSMISSION / RECONCILIATION

• JOBS + EXPORTS

• SUPPLIER / R&D SPILLOVERS - FISCAL COST

DISTRIBUTIONAL / STABILITY LIMIT

• ADDITIONAL INVESTMENT + OUTPUT

• DOMESTIC VALUE + PRODUCTIVITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JOBS + EXPORTS; SUPPLIER / R&D SPILLOVERS - FISCAL COST.

06

STAGE 06

ANCHOR-SUPPLIER BRIDGE

ANCHOR-SUPPLIER BRIDGE

LARGE ANCHOR FIRM

DELIVERY REQUIREMENTS

MSME SUPPLIER UPGRADING

BROADER DOMESTIC SPILLOVERS

LARGE ANCHOR FIRM

→

QUALITY / COST / DELIVERY REQUIREMENTS

→

MSME SUPPLIER UPGRADING

→

BROADER DOMESTIC SPILLOVERS

CONCEPT / ACCOUNTING BASIS

• LARGE ANCHOR FIRM

• QUALITY / COST / DELIVERY REQUIREMENTS

TRANSMISSION / RECONCILIATION

• MSME SUPPLIER UPGRADING

• BROADER DOMESTIC SPILLOVERS

DISTRIBUTIONAL / STABILITY LIMIT

• LARGE ANCHOR FIRM

• QUALITY / COST / DELIVERY REQUIREMENTS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALITY / COST / DELIVERY REQUIREMENTS; MSME SUPPLIER UPGRADING; BROADER DOMESTIC SPILLOVERS.

07

STAGE 07

SEMICONDUCTOR CHAIN

SEMICONDUCTOR CHAIN

DESIGN + IP

MATERIALS + EQUIPMENT

WAFER FABRICATION

DOWNSTREAM ELECTRONICS

MATERIALS + EQUIPMENT

→

WAFER FABRICATION

→

ATMP / OSAT → DOWNSTREAM ELECTRONICS

DESIGN + IP

↓

WAFER FABRICATION

CONCEPT / ACCOUNTING BASIS

• DESIGN + IP

• MATERIALS + EQUIPMENT

TRANSMISSION / RECONCILIATION

• WAFER FABRICATION

• ATMP / OSAT → DOWNSTREAM ELECTRONICS

DISTRIBUTIONAL / STABILITY LIMIT

• DESIGN + IP

• MATERIALS + EQUIPMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WAFER FABRICATION; ATMP / OSAT → DOWNSTREAM ELECTRONICS.

08

STAGE 08

FAB VERSUS PACKAGING

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MANUFACTURES WAFERS

ASSEMBLY + TEST + MARK + PACKAGE

ARCHITECTURE + IP

FINAL ELECTRONICS; NOT AUTOMATICALLY A FAB

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
FAB → manufactures wafers	DESIGN → architecture + IP	FAB → manufactures wafers
ATMP / OSAT → assembly + test + mark + package	ASSEMBLY → final electronics; not automatically a fab	ATMP / OSAT → assembly + test + mark + package

MEASURE / INSTITUTION

• FAB → manufactures wafers

• ATMP / OSAT → assembly + test + mark + package

DECISIVE DISTINCTION

• DESIGN → architecture + IP

• ASSEMBLY → final electronics; not automatically a fab

VINTAGE / LEGAL LIMIT

• FAB → manufactures wafers

• ATMP / OSAT → assembly + test + mark + package

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ASSEMBLY → final electronics; not automatically a fab.

09

STAGE 09

FAB ENABLING ECOSYSTEM

FAB ENABLING ECOSYSTEM

RELIABLE POWER + ULTRA-PURE WATER

PROCESS TALENT + TECHNOLOGY PARTNER

CLEAN LOGISTICS + MATERIALS

DEMAND + SUPPLIERS + LONG GESTATION

RELIABLE POWER + ULTRA-PURE WATER

→

PROCESS TALENT + TECHNOLOGY PARTNER

→

CLEAN LOGISTICS + MATERIALS

→

DEMAND + SUPPLIERS + LONG GESTATION

CONCEPT / ACCOUNTING BASIS

• RELIABLE POWER + ULTRA-PURE WATER

• PROCESS TALENT + TECHNOLOGY PARTNER

TRANSMISSION / RECONCILIATION

• CLEAN LOGISTICS + MATERIALS

• DEMAND + SUPPLIERS + LONG GESTATION

DISTRIBUTIONAL / STABILITY LIMIT

• RELIABLE POWER + ULTRA-PURE WATER

• PROCESS TALENT + TECHNOLOGY PARTNER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLEAN LOGISTICS + MATERIALS; DEMAND + SUPPLIERS + LONG GESTATION

- FAB → manufactures wafers
- ATMP / OSAT → assembly + test + mark + package

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10

STAGE 10 DOMESTIC-VALUE TRAP

DOMESTIC-VALUE TRAP

GROSS OUTPUT RISES

BUT HIGH-VALUE INPUTS MAY REMAIN IMPORTED

SHALLOW LOCAL VALUE ADDITION

MEASURE STAGE-WISE CAPABILITY

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MEASURE stage-wise capability

CONCEPT / ACCOUNTING BASIS

- GROSS OUTPUT RISES
- BUT HIGH-VALUE INPUTS MAY REMAIN IMPORTED

TRANSMISSION / RECONCILIATION

- SHALLOW LOCAL VALUE ADDITION
- MEASURE stage-wise capability

DISTRIBUTIONAL / STABILITY LIMIT

- GROSS OUTPUT RISES
- BUT HIGH-VALUE INPUTS MAY REMAIN IMPORTED

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUT HIGH-VALUE INPUTS MAY REMAIN IMPORTED; SHALLOW LOCAL VALUE ADDITION; MEASURE stage-wise capability.

11

STAGE 11 MANUFACTURING ANSWER SPINE

MANUFACTURING ANSWER SPINE

CLASSIFY FIRM + BIND CONSTRAINT

MATCH INSTRUMENT TO FINANCE

TRACE SEMICONDUCTOR STAGES AND STATUS

EVALUATE ADDITIONALITY + SPILLOVERS + EXIT RULES

CLASSIFY firm + bind constraint

MATCH instrument to finance / scale / technology

TRACE semiconductor stages and status

EVALUATE additionality + spillovers + exit rules

CONCEPT / ACCOUNTING BASIS

- CLASSIFY firm + bind constraint
- MATCH instrument to finance / scale / technology

TRANSMISSION / RECONCILIATION

- TRACE semiconductor stages and status
- EVALUATE additionality + spillovers + exit rules

DISTRIBUTIONAL / STABILITY LIMIT

- CLASSIFY firm + bind constraint
- MATCH instrument to finance / scale / technology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MATCH instrument to finance / scale / technology.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

MSME INFORMALITY CAN BE AN ADAPTATION TO COMPLIANCE AND

CLUSTER POLICY LOWERS SHARED COSTS FOR TESTING, DESIGN, SKILLING

PLI ADDITIONALITY ASKS WHETHER OUTPUT, INVESTMENT OR TECHNOLOGY WOULD

SEMICONDUCTOR SUPPORT MUST MANAGE LONG GESTATION, TECHNOLOGY OBSOLESCENCE AND

DOMESTIC VALUE ADDITION SHOULD BE MEASURED ACROSS STAGES, NOT

SUPPLIER DEVELOPMENT AND R&D LINKS DETERMINE WHETHER AN ANCHOR

MINISTRY OF MSME AND SIDBI

OPTIONAL MODEL / MEASUREMENT DEPTH

- CAUTION: MSME informality can be an adaptation to compliance and finance constraints, not simply
- CAUTION: Cluster policy lowers shared costs for testing, design, skilling, effluent treatment
- CAUTION: PLI additionality asks whether output, investment or technology would have occurred
- CAUTION: Semiconductor support must manage long gestation, technology obsolescence and global

VINTAGE, BASKET OR LEGAL LIMIT

- CAUTION: Domestic value addition should be measured across stages, not inferred from final
- CAUTION: Supplier development and R&D links determine whether an anchor investment generates
- Ministry of MSME and SIDBI: support enterprise development, credit and cluster
- RBI-regulated TReDS platforms: facilitate discounting of accepted MSME receivables.

COMPARATIVE POLICY NUANCE

- Line ministries administering PLI schemes: verify sector-specific investment and
- India Semiconductor Mission and MeitY: coordinate semiconductor ecosystem support.
- CAUTION: TReDS finances an accepted invoice; it is not collateral-based machinery finance or a
- CAUTION: Final assembly can expand rapidly while domestic value addition remains low if high

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.